

Correction Note — Step 7: Training Seed Robustness (Proposed Revisions for Final Report)

1. Clarification of Downstream Evaluation Coverage

While five models were trained using different random seeds, the original report explicitly presents downstream simulation comparison only for the best and worst seeds. To align the claims with the full experiment, we include a consolidated summary across all five seeds.

Seed	Val MSE	λ (/yr)	Final RMS (AU)	Bounded	Speedup
42	1.11e-5	0.1655	1.54	YES	1.07×
123	1.72e-5	0.1655	—	YES	—
456	1.75e-5	0.1655	1.23	YES	1.07×
789	1.68e-5	0.1655	—	YES	—
1000	1.43e-5	0.1655	—	YES	—

2. Revised Interpretation of Seed Sensitivity

Training MSE exhibits moderate variability across seeds (CV $\approx$ 15.8%), but all five independently trained models produce identical divergence rates and bounded trajectories over the evaluation horizon. This indicates that simulation behavior is robust to model initialization.

3. Mechanistic Explanation

The observed MSE variation is concentrated at sub-softening distances where the safety gate bypasses neural network predictions. As a result, these variations do not affect the simulation, making the system robust by design rather than by chance.

4. Wording Refinement

Replace 'results are fully reproducible for all seeds' with: 'all five independently trained models produce identical divergence rates within numerical precision and bounded trajectories.'

5. Notation Fixes

Replace corrupted symbols with standard notation: 10^{-5} , 1×10^{-3} , and ensure consistent formatting throughout.

